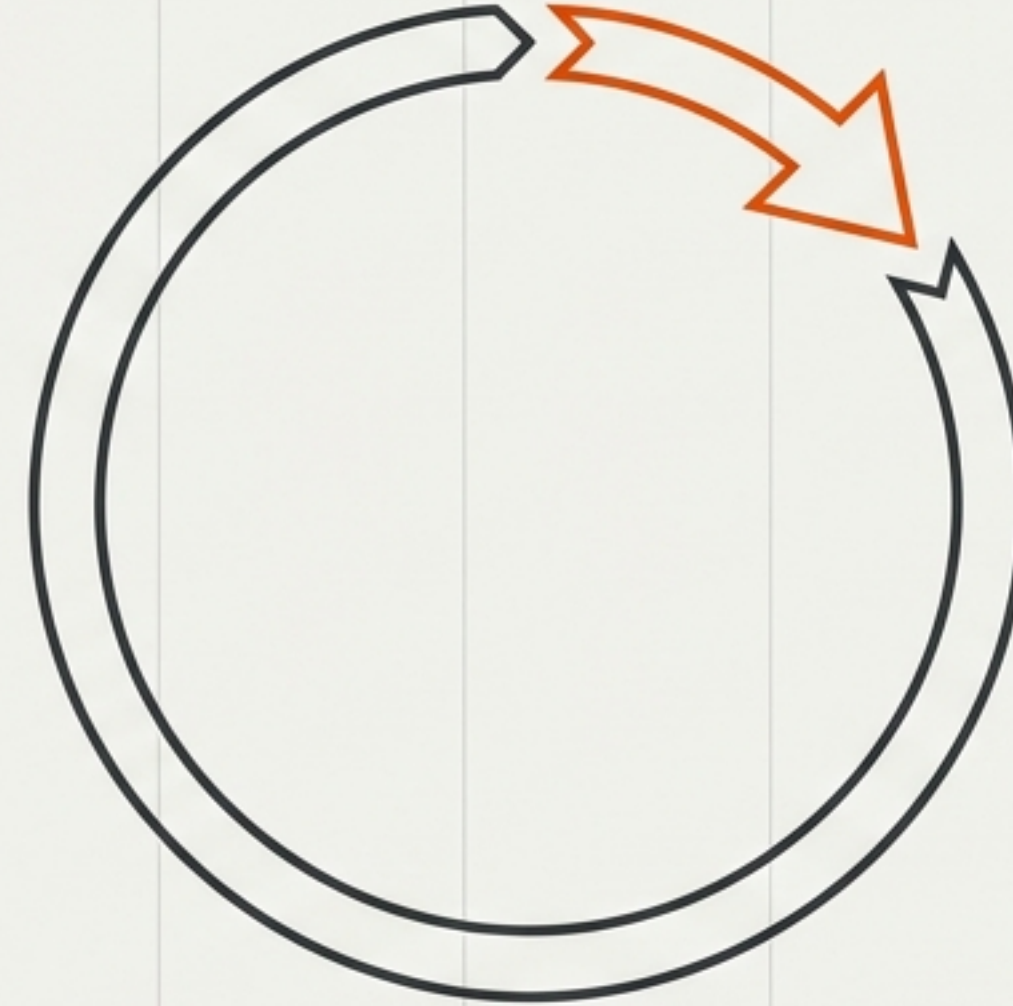


Temporal Delay Acts as a Spatial Boundary

How circulation time alone enforces spectral discreteness in feedback systems.



The Paradigm Shift

Traditional physics assumes mode discretization requires spatial boundary conditions (like a string fixed at both ends). This research demonstrates that finite circulation delay (τ) is sufficient to enforce discrete modes without any spatial structure.

The Mechanism

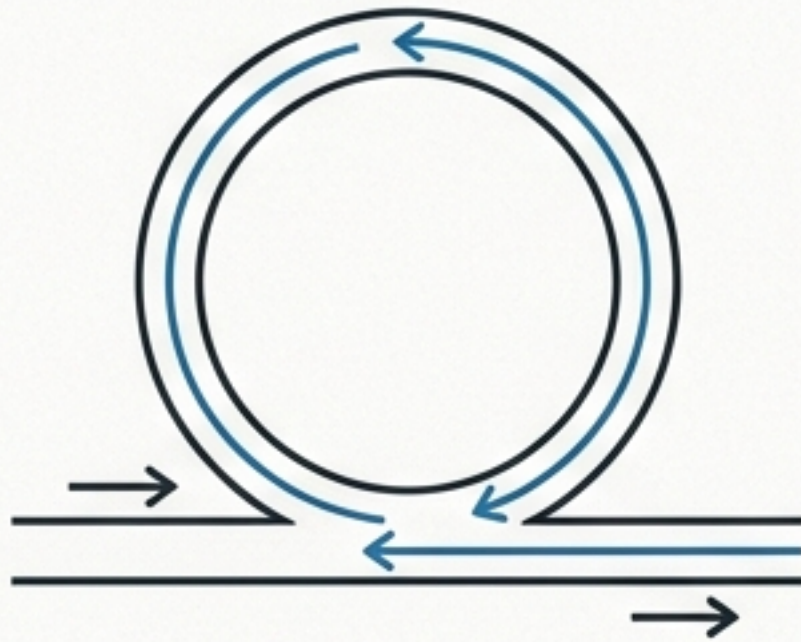
Finite transit time acts as a "selection filter," stabilizing specific frequencies through dynamical stability constraints.

The Implication

Delay-induced discreteness is a universal feature of effective field theories, parallel to how hidden microstructure creates new properties in metamaterials.

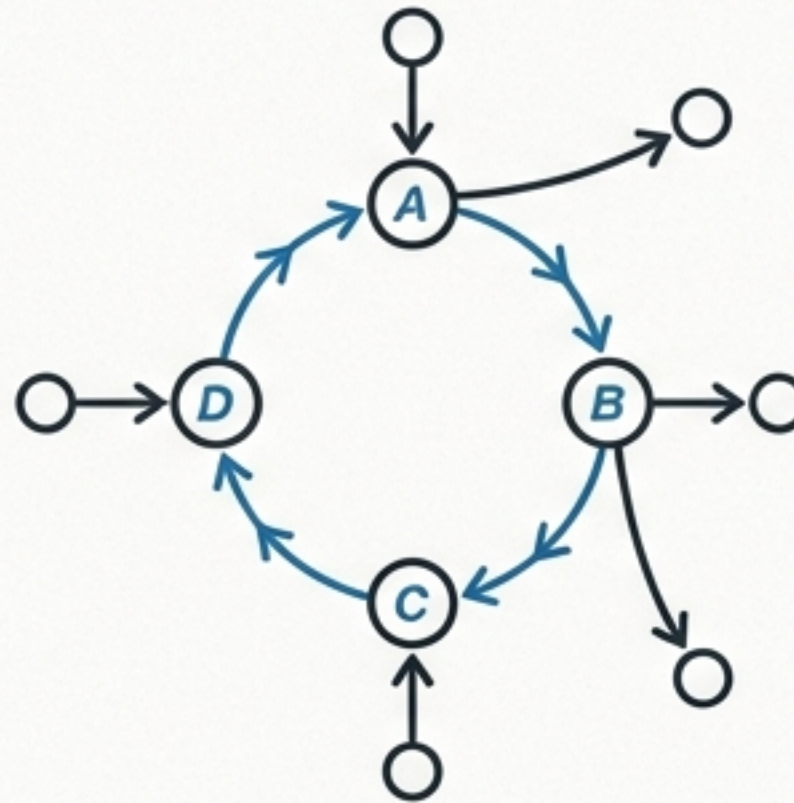
Closed Loops Are Everywhere

Closed-loop systems with finite signal propagation times are generic across physics, engineering, and biology. Despite their differences, these systems share a common trait: propagation time is comparable to the intrinsic timescale of the field dynamics.



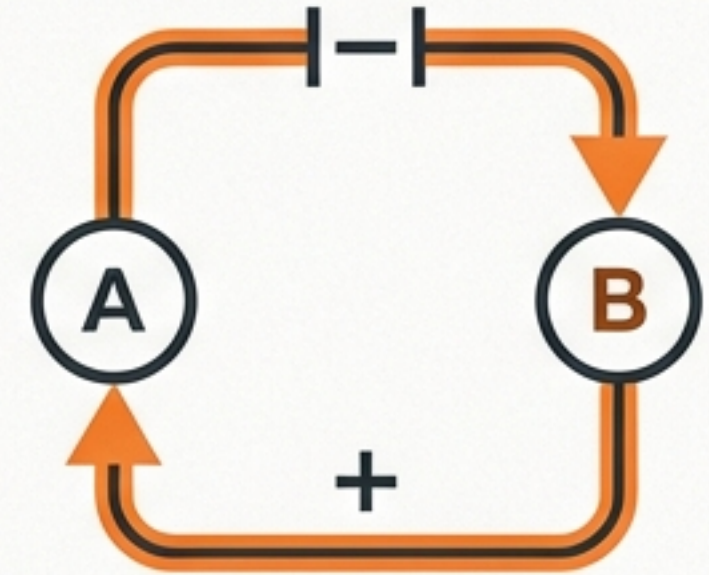
Optical & Electronic

Optical ring resonators,
phase-locked loops.



Network Dynamics

Delayed neural networks,
control systems.

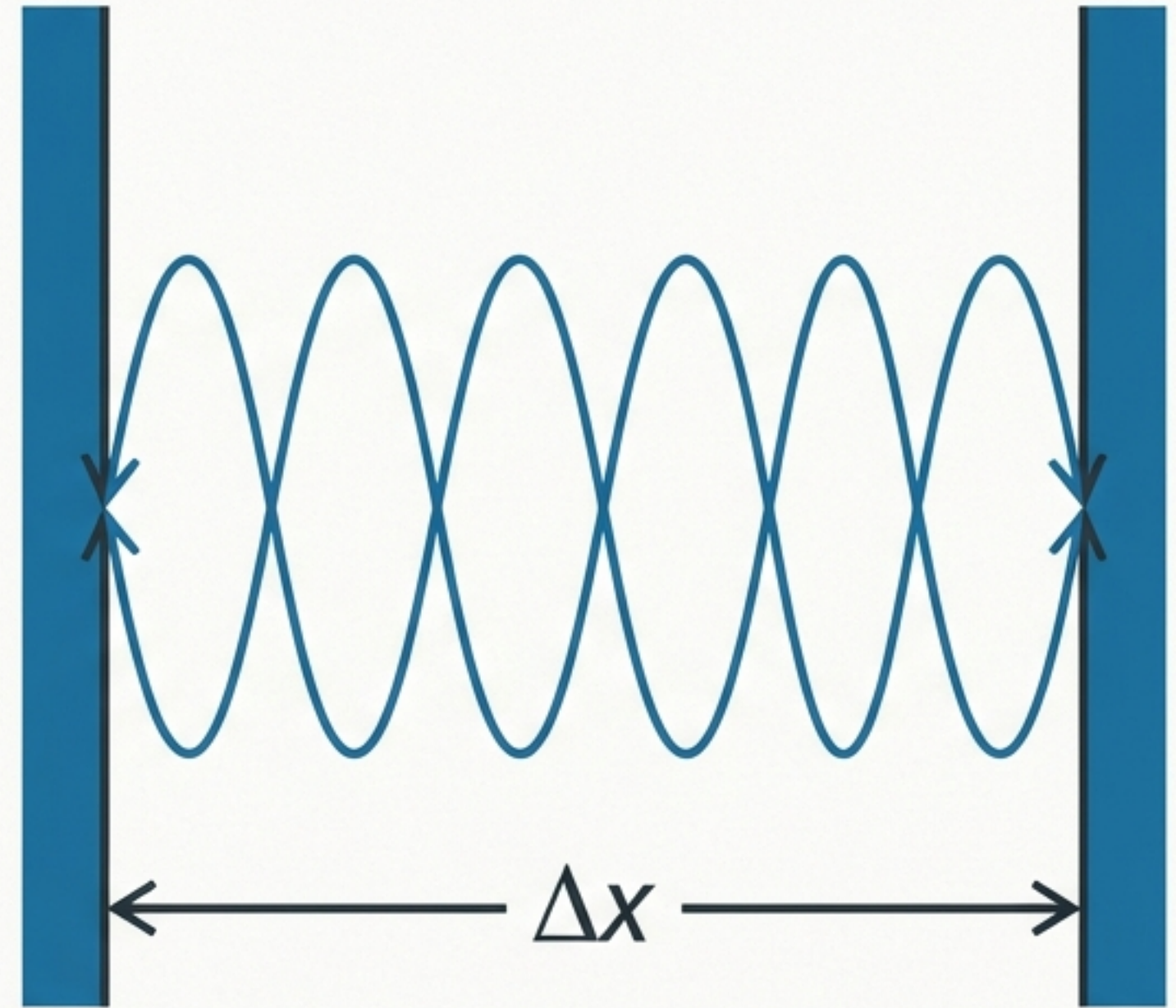


Biological

Feedback loops in regulatory
systems.

The Traditional View: Geometry Dictates Structure

- In classical field theories, modes are defined by spatial boundary conditions.
- Think of a guitar string or a particle in a box: the physical constraints of the geometry force the wave to take a standing form.
- Mathematical convention: Local constitutive relations where fluxes depend on instantaneous, local gradients.



Standard Model: Spatial constraints (Δx) enforce discreteness.

The Shift: **Time** as the Fundamental Constraint

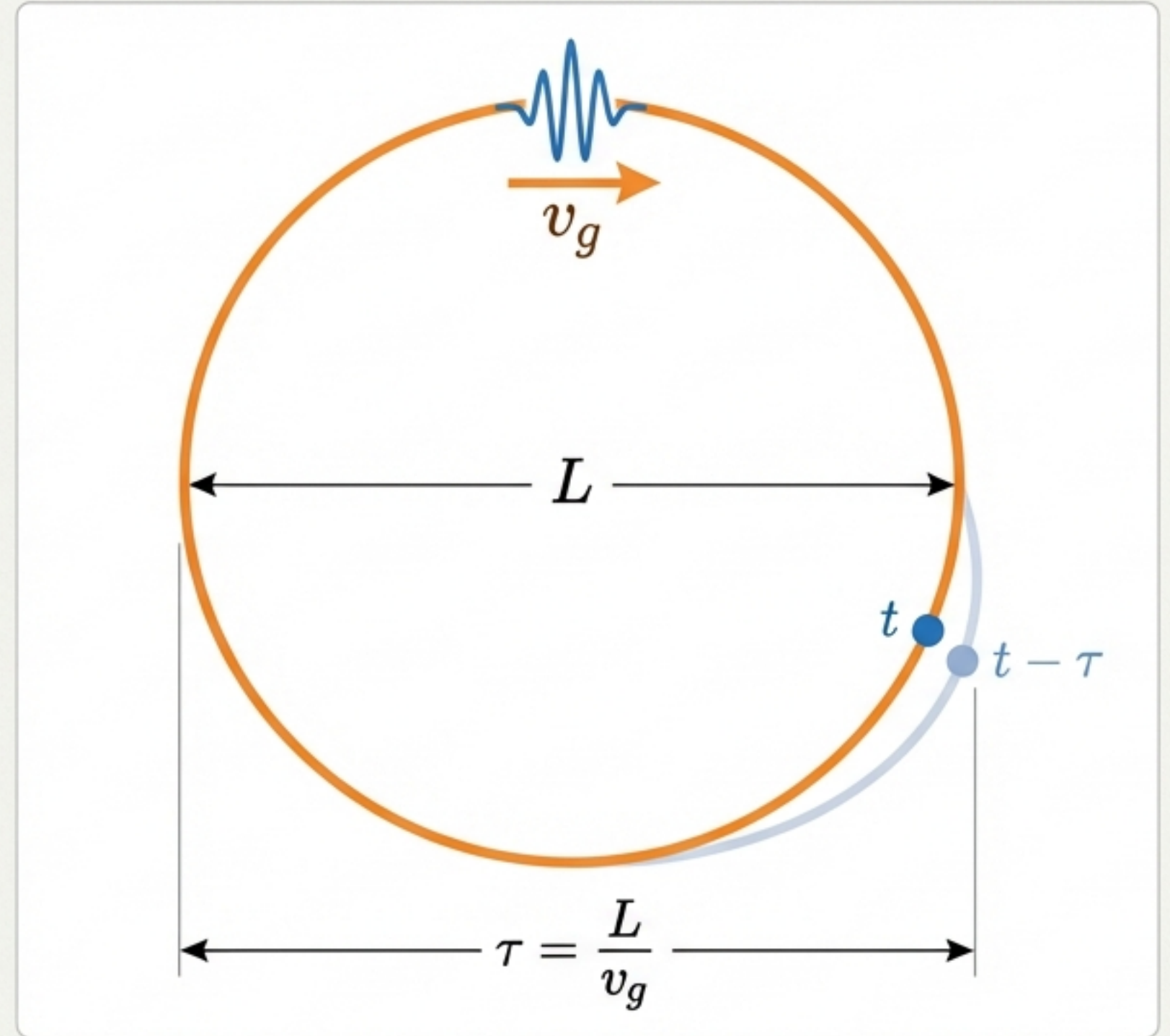
- What happens if we remove the spatial boundaries? Does the system become chaotic?
- **The Thesis:** Even in the absence of spatial structure, temporal circulation delay acts as an effective constitutive ingredient.
- **Key Definition:** Circulation Time $\tau = L/v_g$ (where L is effective length and v_g is group velocity).



The finite transit time introduces "**temporal nonlocality**," fundamentally modifying the system's spectral properties.

The Physical Framework

- **The Setup:** A scalar excitation propagates on a closed feedback loop.
- **The Condition:** We operate where propagation time is significant.
- **The Shift:** We move from static spatial boundaries to a dynamic perspective: the signal interacts with its own past self after one rotation.



Feedback Loop Model: A scalar field interacts with its own past state after circulating a closed path of length L with group velocity v_g .

The Minimal Phase Model

Isolating the phase degree of freedom to expose the minimal ingredients required for discreteness. Crimson Pro.

$$\dot{\phi}(t) = \omega_0 + \kappa \sin[\phi(t - \tau) - \phi(t)]$$

**Instantaneous
phase evolution**

**Intrinsic
Frequency**
(Natural speed
of oscillator)

**Feedback Coupling
Strength**

Delayed Self-Interaction
The system remembers its
state from one loop ago.

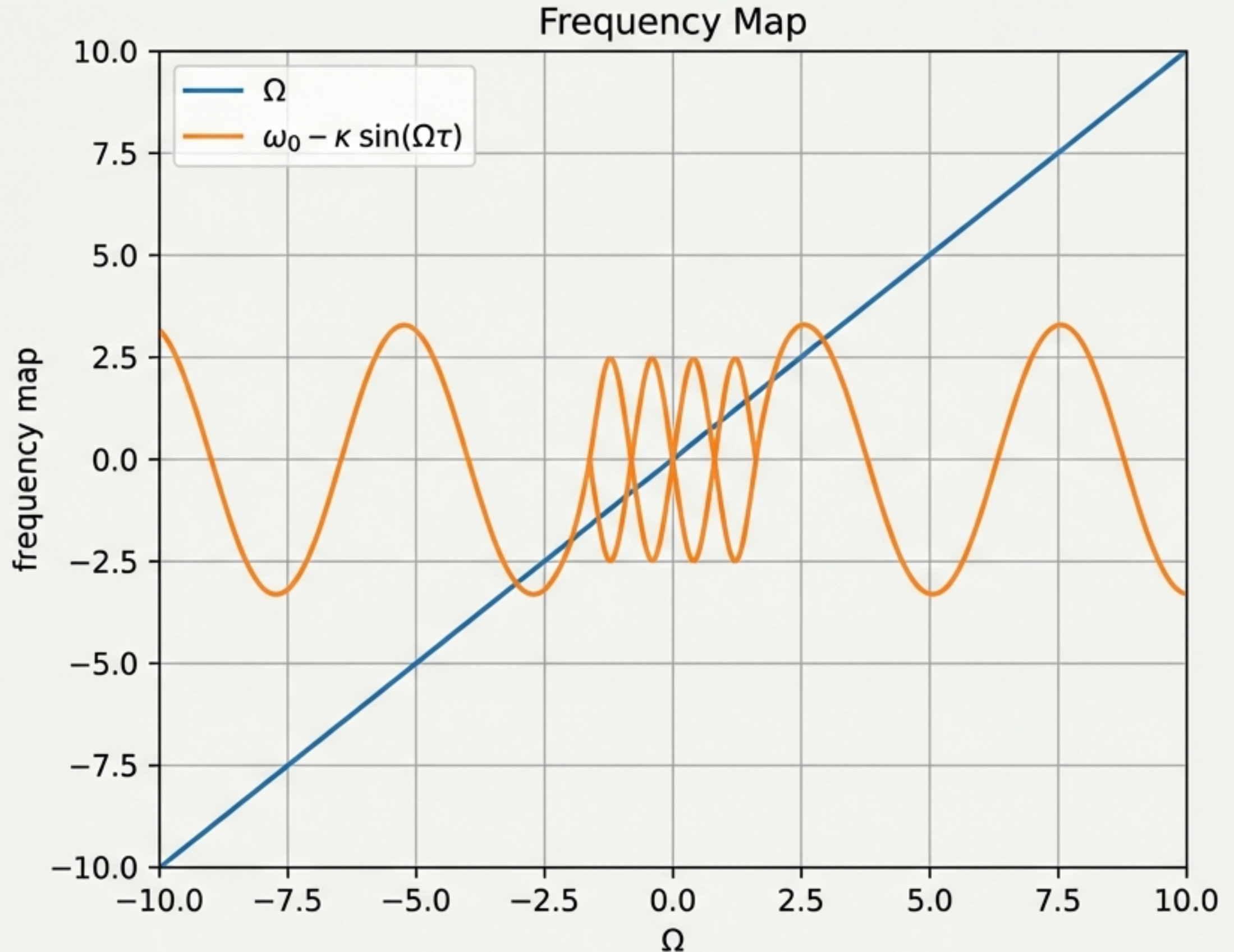
Emergent Spectral Discreteness

We seek uniformly rotating solutions: $\phi(t) = \Omega t + \phi_0$.

This yields the transcendental condition:

$$\Omega = \omega_0 - \kappa \sin(\Omega\tau)$$

The continuous line is broken by the delay term. Every intersection point represents a potential mode.



The Stability Filter

Nature performs a 'survival of the fittest' on frequencies.
Not every intersection point is stable.

Linearizing about a locked solution yields the variational equation:

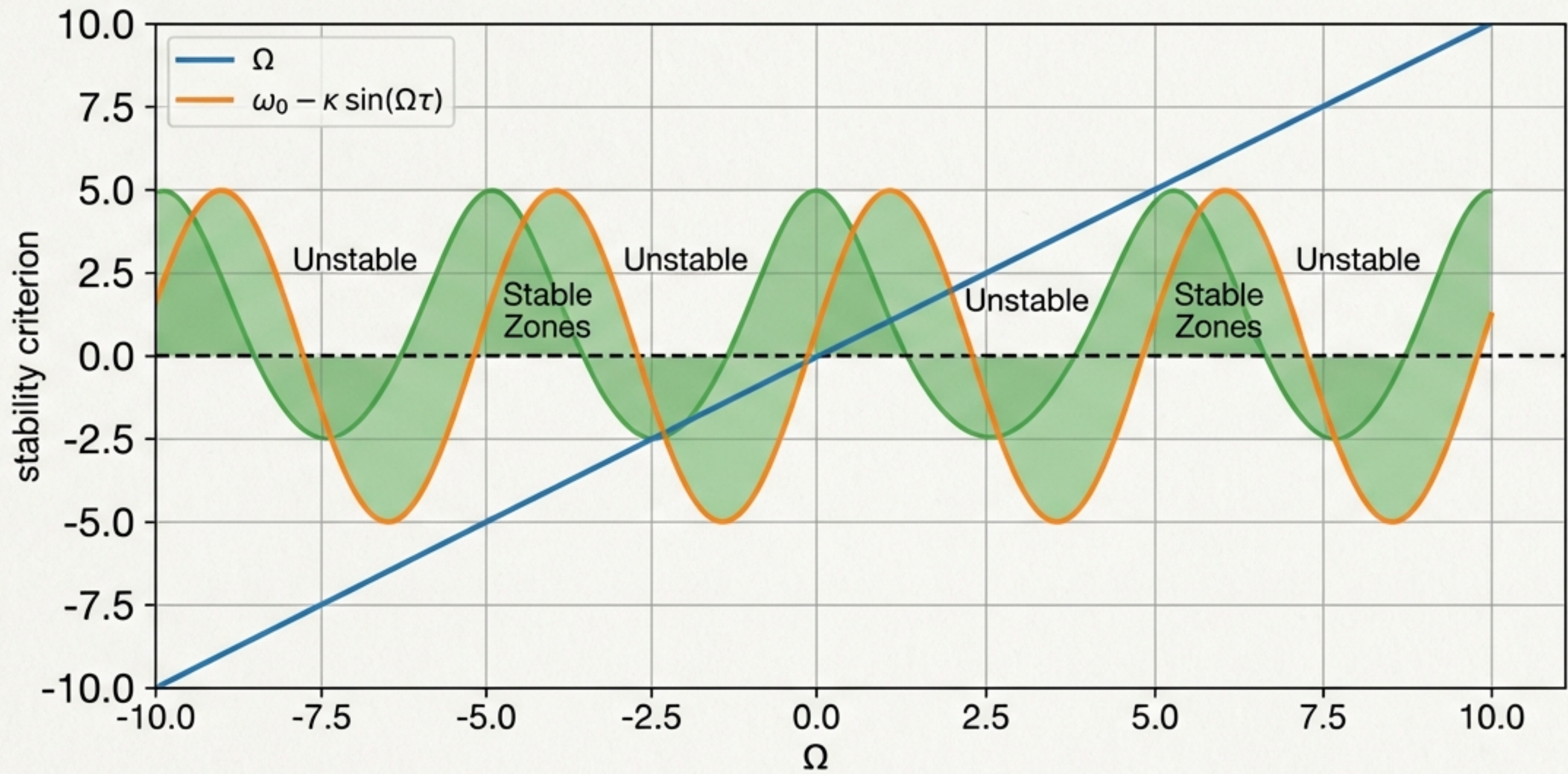
$$\dot{\psi}(t) = -\gamma\psi(t) - \gamma\psi(t - \tau)$$

The Selection Rule:

$$1 + \kappa\tau \cos(\Omega\tau) > 0$$

This inequality acts as a dynamical selection rule. It suppresses unstable branches and enforces robust operating states.

Visualizing Natural Selection of Modes



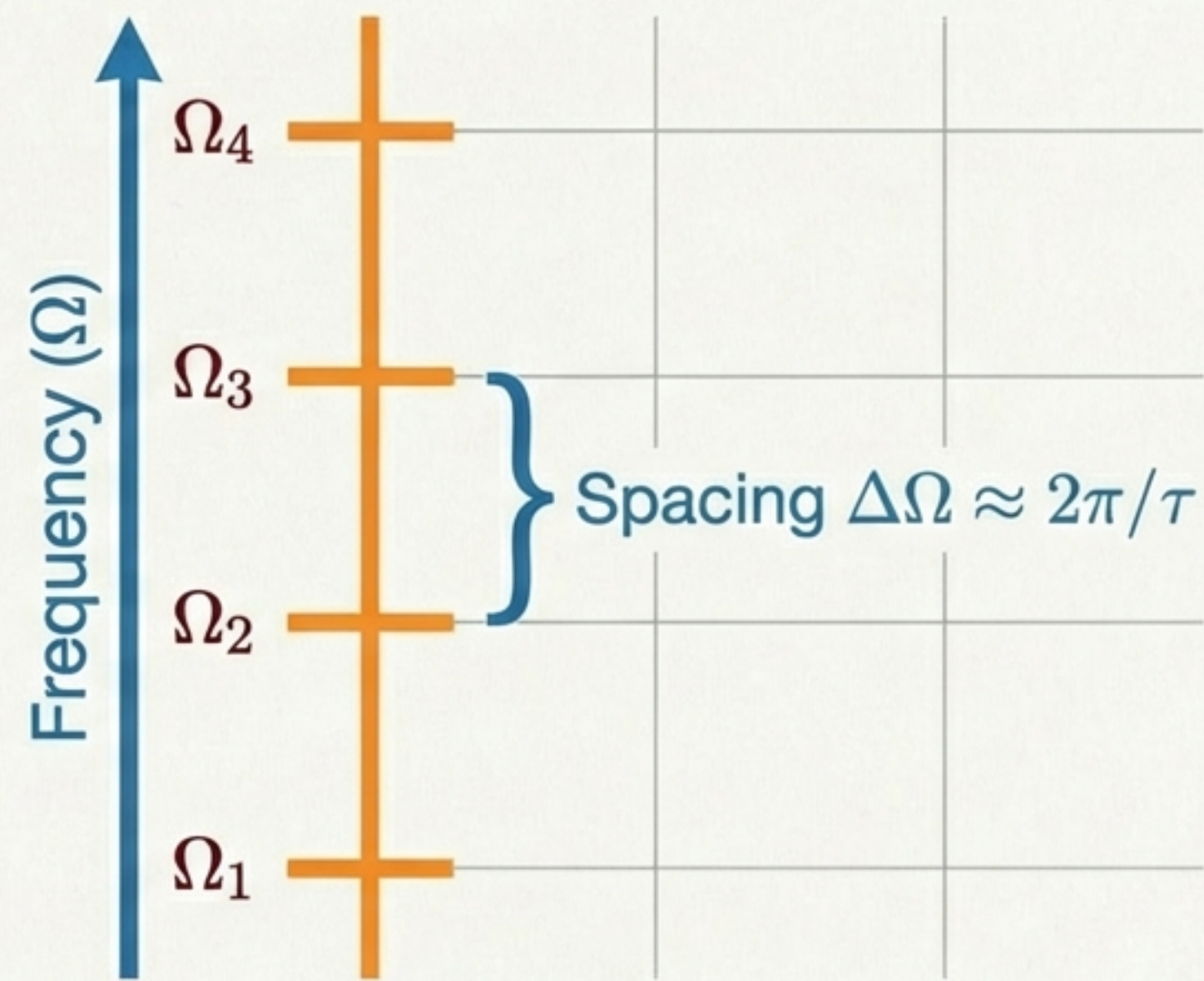
The Result: A Discrete Ladder

In the long-delay regime ($\kappa\tau \gg 1$), stable solutions organize into a discrete ladder:

$$\Omega_n \approx \frac{2\pi n}{\tau} + \delta\Omega, \quad n \in \mathbb{Z}$$

Interpretation: The integer n counts the number of phase windings accumulated during one circulation time τ . This is analogous to a topological winding number.

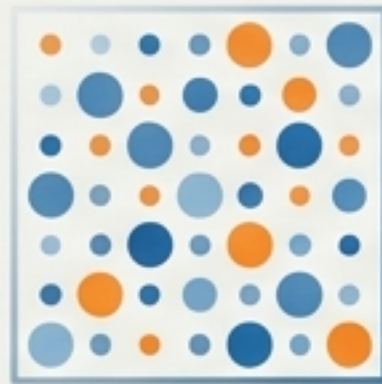
The Frequency Ladder



The Metamaterial Analogy

Bianisotropic Homogenization

In Metamaterials: Hidden spatial microstructure creates effective properties (like coupling heat flux directly to temperature).



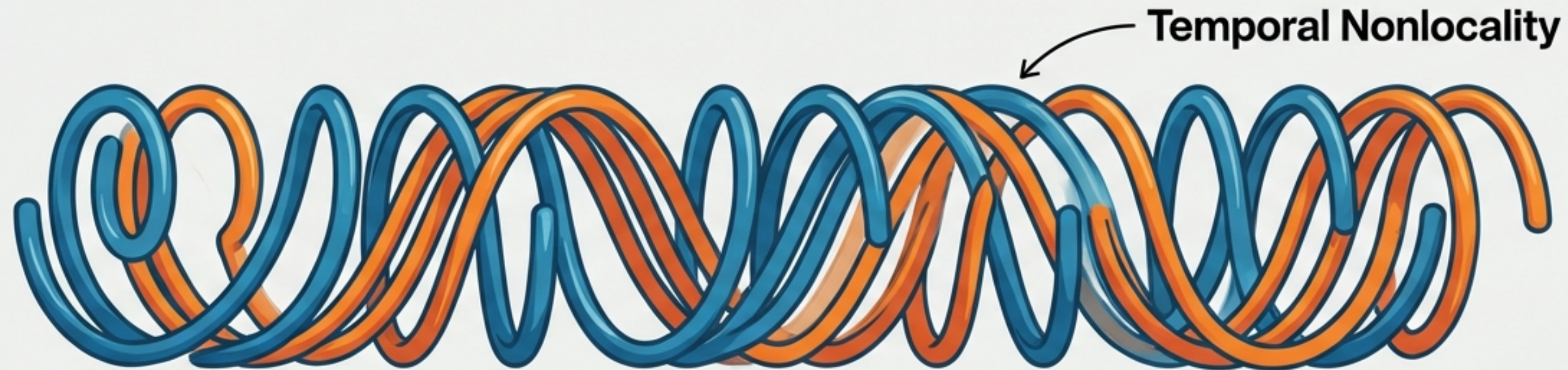
Temporal Homogenization

In Feedback Loops: Hidden temporal history necessitates an expanded macroscopic description.



“Just as spatial homogenization reveals hidden constitutive structure, the history dependence of a circulating field generates discrete, stability-selected operating states.”

Memory as a Constitutive Ingredient



Multistability is often treated as a secondary feature in control theory.

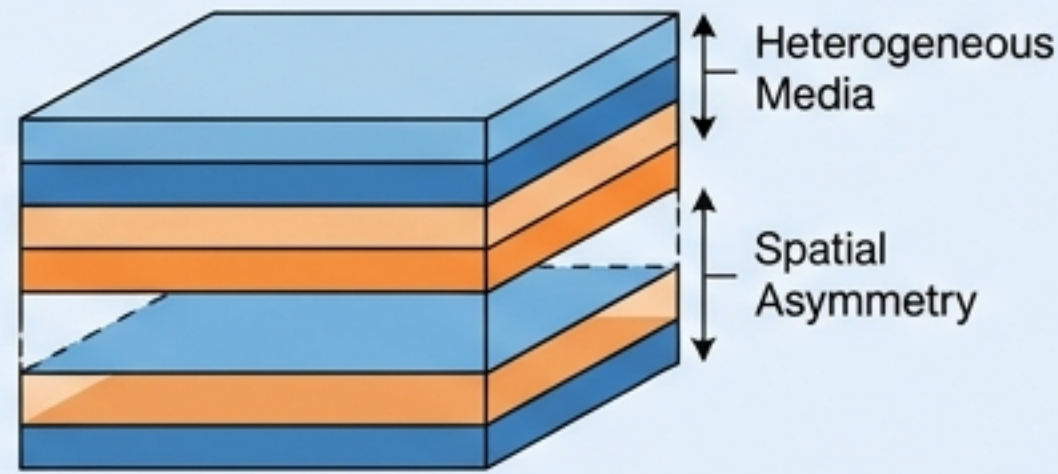
New Perspective: We reinterpret it as an emergent discretization mechanism.

Temporal nonlocality (memory) plays the role of an effective constitutive ingredient, legitimizing additional dynamical structure at the macroscopic level without needing microscopic quantization.

Space vs. Time: Two Paths to Structure

SPACE

Spatial Homogenization

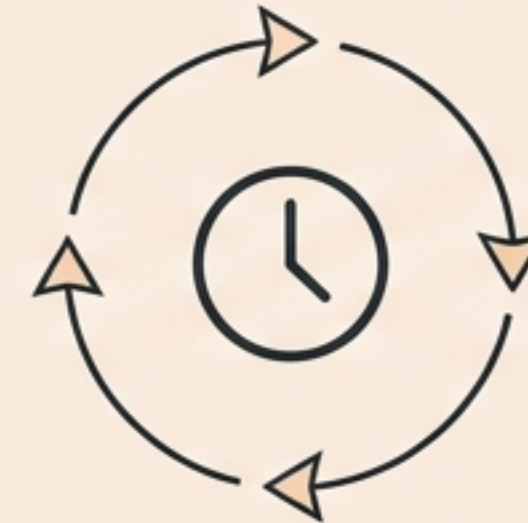


Mechanism: Spatial asymmetry in heterogeneous media.

Result: Emergent effective couplings.

TIME

Temporal Nonlocality



Mechanism: Delayed feedback in circulation loops.

Result: Emergent spectral discreteness.

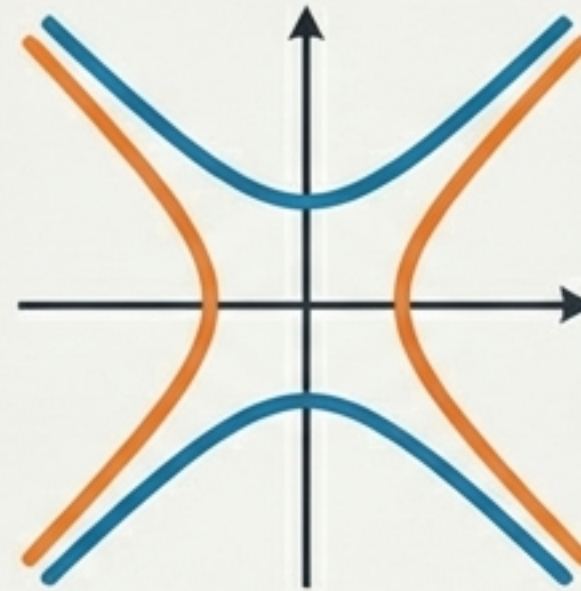
Key Takeaway: Both generate emergent effective structure.

Resolving Theoretical Paradoxes



Avoids Infinite Speed

By explicitly incorporating finite circulation time, the model avoids the infinite-speed paradox associated with parabolic transport theories.



Hyperbolic Features

The resulting dynamics exhibit hyperbolic-like features, akin to "second-sound" phenomena in nonlocal thermal media.



Universal Feature

Suggests that delay-induced discreteness is a generic feature of effective field theories in loops.

Conclusion: Time is Sufficient for Order



1. We successfully derived a minimal phase-oscillator model for a circulating field.
2. We demonstrated that finite transit time acts as a selection filter, stabilizing a discrete ladder of modes.
3. This occurs without spatial standing-wave boundary conditions or microscopic quantization.

Temporal nonlocality is a generative mechanism for emergent macroscopic structure. We don't need boxes to have order; we just need time.